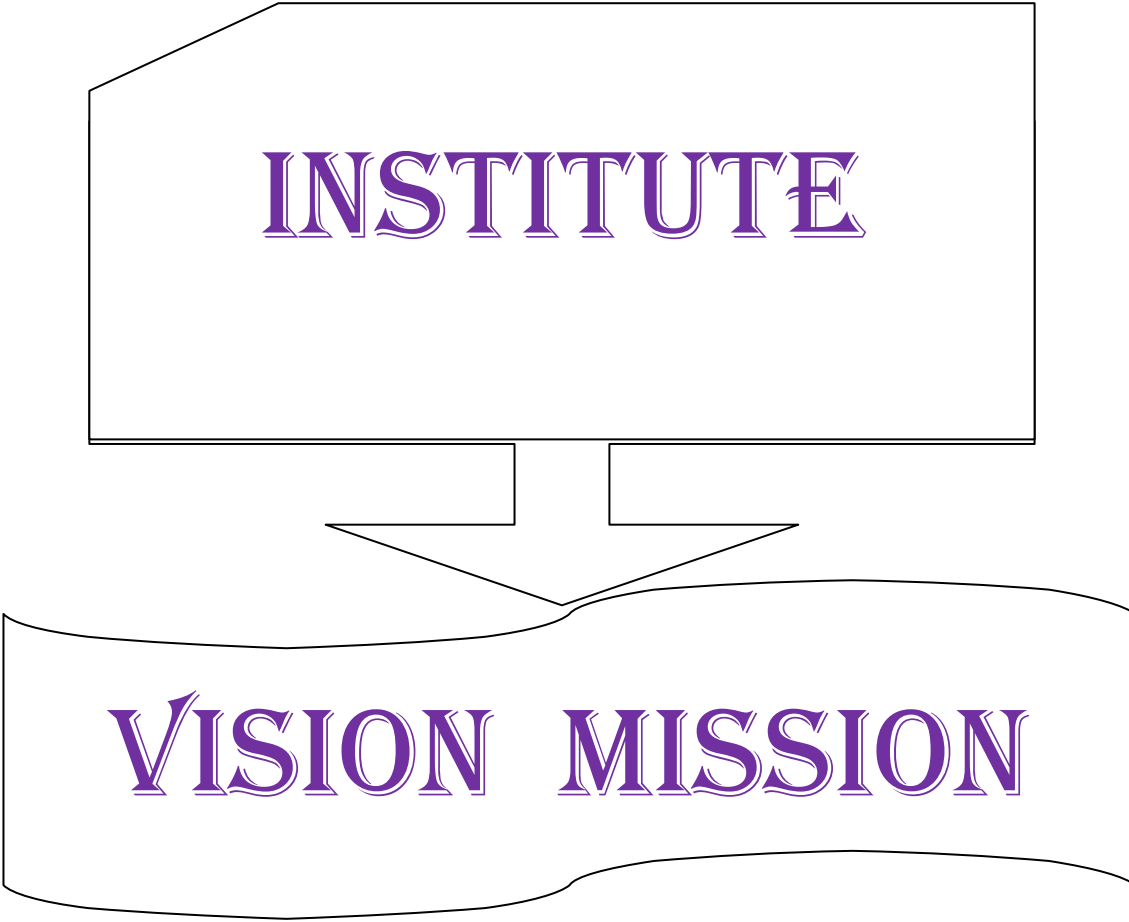


Academic Year  
2022-23  
**III SEM CSE**  
**Handbook**  
(V20 Regulation)

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INSTITUTE

VISION MISSION

# INSTITUTE VISION AND MISSION

## VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

## MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

# DEPARTMENT

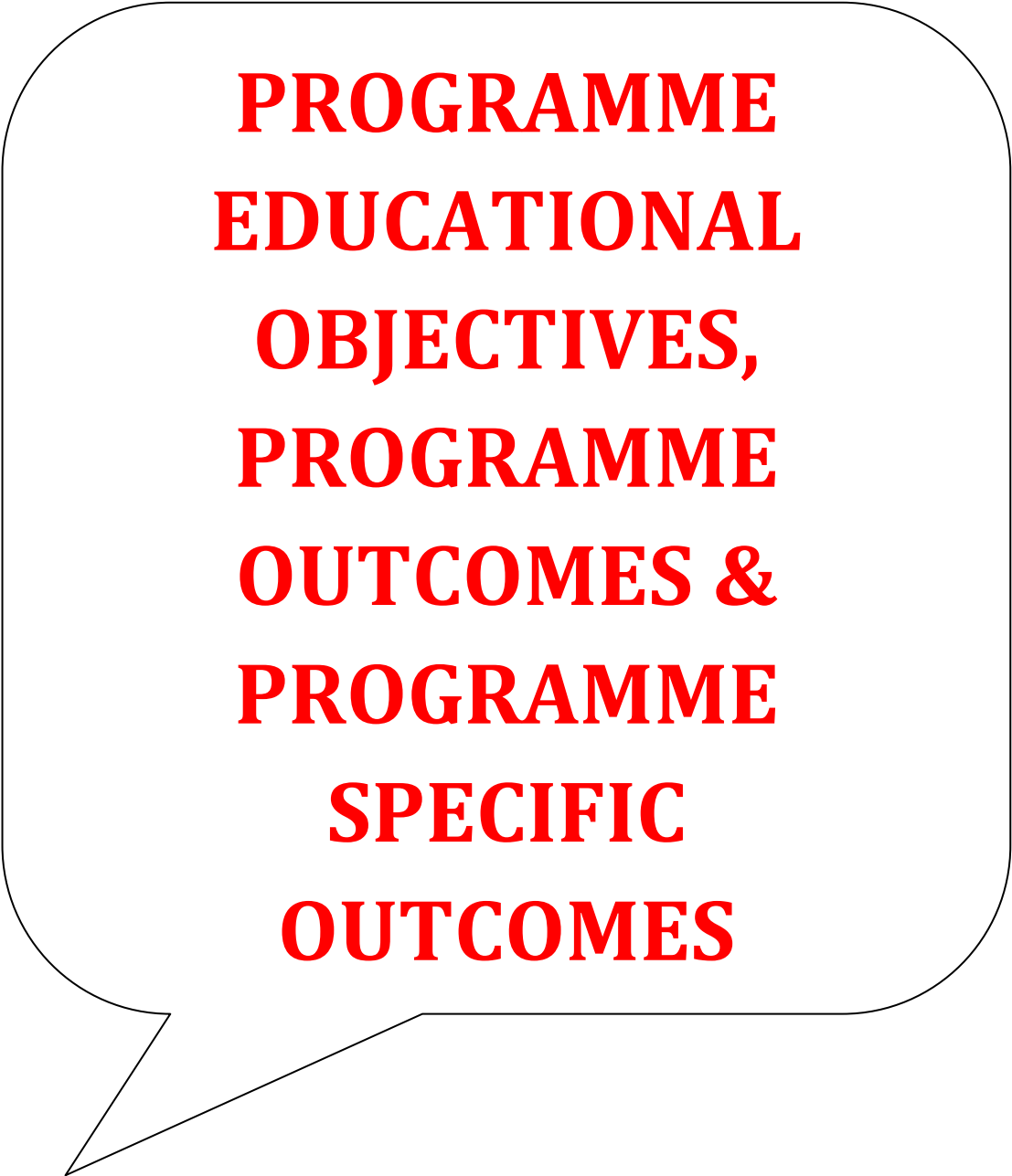
## VISION AND MISSION

### **Vision:**

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

### **Mission :**

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character



**PROGRAMME  
EDUCATIONAL  
OBJECTIVES,  
PROGRAMME  
OUTCOMES &  
PROGRAMME  
SPECIFIC  
OUTCOMES**

## **Programme Educational Objectives (PEOs)**

Graduates of this programme will :

**PEO 1:** Adapt to evolving technology.

**PEO 2:** Provide optimal solutions to real time problems.

**PEO 3:** Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

## **Programme Specific Outcomes (PSOs):**

A graduate of the Computer Science and Engineering Program will be able to:

**PSO 1:** Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

**PSO 2:** Use Professional engineering practices and strategies for development and maintenance of software. [K3]



## Programme Outcomes (POs):

### Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. [K3]
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. [K4]
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. [K5]
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [K5]
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. [K3]
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. [K3]
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. [K3]
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. [K3]
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [K6]
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. [K2]
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [K6]
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [K1]

# ACADEMIC CALENDAR

✉ : [principal@srivasaviengg.ac.in](mailto:principal@srivasaviengg.ac.in)  
[svec.a8@gmail.com](mailto:svec.a8@gmail.com)



☎ : 08818- 284344, 355

## **SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)**

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)

Pedatadepalli, TADEPALLIGUDEM – 534 101. W.G.Dist. (A.P)

Principal's Office

Date: 17-09-2022

### **Academic Calendar** **For B.Tech III and IV Semesters, Academic Year 2022-23**

| III Semester                           |            |            |       |
|----------------------------------------|------------|------------|-------|
| Description                            | From       | To         | Weeks |
| Commencement of Class Work             | 03.10.2022 |            |       |
| I Unit of Instructions                 | 03.10.2022 | 19.11.2022 | 7 W   |
| I Mid Examinations                     | 21.11.2022 | 26.11.2022 | 1 W   |
| II Unit of Instructions                | 28.11.2022 | 14.01.2023 | 7 W   |
| II Mid Examinations                    | 16.01.2023 | 21.01.2023 | 1 W   |
| Preparation & Practicals               | 23.01.2023 | 28.01.2023 | 1 W   |
| End Examinations                       | 30.01.2023 | 11.02.2023 | 2 W   |
| Commencement of IV Semester Class Work | 13.02.2023 |            |       |
| IV Semester                            |            |            |       |
| Description                            | From       | To         | Weeks |
| Commencement of Class Work             | 13.02.2023 |            |       |
| I Unit of Instructions                 | 13.02.2023 | 01.04.2023 | 7 W   |
| I Mid Examinations                     | 03.04.2023 | 08.04.2023 | 1 W   |
| II Unit of Instructions                | 10.04.2023 | 27.05.2023 | 7 W   |
| II Mid Examinations                    | 29.05.2023 | 03.06.2023 | 1 W   |
| Preparation & Practicals               | 05.06.2023 | 10.06.2023 | 1 W   |
| End Examinations                       | 12.06.2023 | 23.06.2023 | 2 W   |

  
PRINCIPAL

Copy to : ALL

#### **Vision**

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

#### **Mission**

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



# SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-584 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



## CLASS CONSOLIDATED TIME TABLE

Class: III SEM

Section - A

Class Coordinator: Mr. P Umasankar

Room: B-301

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                  | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | MEFA                    | MFCS                    | COA                     | OOPS                    | LUNCH<br>BREAK   | LSS LAB                 |                         |                         |
| Tue         | DS                      | OOPS LAB                |                         |                         |                  | MFCS                    | PCS-I                   | COA                     |
| Wed         | MFCS                    | MEFA                    | OOPS                    | LIBRARY                 |                  | DS                      | MEFA                    | OOPS                    |
| Thu         | DS                      | DS LAB                  |                         |                         |                  | OOPS                    | COA                     | MFCS                    |
| Fri         | OOPS                    | DS                      | LSS                     | COA                     |                  | MFCS                    | MEFA                    | SPORTS                  |
| Sat         | COA                     | DS                      | DS                      | MEFA                    |                  | PCS-I                   | PCS-I                   | OOPS                    |

Class: III SEM

Section - B

Class Coordinator: Mr. K Lakshmaji

Room: B-302

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM       | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                        | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | COA                     | OOPS                    | DS                      | MFCS                    | <b>LUNCH<br/>BREAK</b> | <b>DS LAB</b>           |                         |                         |
| Tue         | MFCS                    | MEFA                    | DS                      | COA                     |                        | DS                      | OOPS                    | <b>SPORTS</b>           |
| Wed         | DS                      | MEFA                    | MFCS                    | LSS                     |                        | OOPS                    | DS                      | <b>LIBRARY</b>          |
| Thu         | COA                     | OOPS                    | COA                     | MEFA                    |                        | <b>LSS LAB</b>          |                         |                         |
| Fri         | MEFA                    | COA                     | OOPS                    | <b>MFCS</b>             |                        | PCS-I                   | PCS-I                   | OOPS                    |
| Sat         | <b>OOPS LAB</b>         |                         |                         | PCS-I                   |                        | MEFA                    | MFCS                    | DS                      |

Class: III SEM

Section - C

Class Coordinator: Mr. K Lakshmi Narayana

Room: B-303

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                  | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | OOPS                    | MFCS                    | COA                     | MEFA                    | LUNCH<br>BREAK   | PCS-I                   | DS                      | SPORTS                  |
| Tue         | COA                     | DS                      | MFCS                    | LIBRARY                 |                  | MEFA                    | OOPS                    | DS                      |
| Wed         | MEFA                    | OOPS                    | PCS-I                   | PCS-I                   |                  | MFCS                    | DS                      | COA                     |
| Thu         | COA                     | MFCS                    | DS                      | OOPS                    |                  | OOPS LAB                |                         |                         |
| Fri         | MFCS                    | LSS                     | OOPS                    | MEFA                    |                  | DS LAB                  |                         |                         |
| Sat         | OOPS                    | LSS LAB                 |                         |                         |                  | DS                      | COA                     | MEFA                    |

Class: III SEM

Section - D

Class Coordinator: Mrs. B Sri Ramya

Room: B-304

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM       | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                        | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | COA                     | <b>DS LAB</b>           |                         |                         | <b>LUNCH<br/>BREAK</b> | MEFA                    | MFCS                    | OOPS                    |
| Tue         | MEFA                    | LSS                     | OOPS                    | DS                      |                        | <b>LSS LAB</b>          |                         |                         |
| Wed         | DS                      | COA                     | MFCS                    | MEFA                    |                        | <b>OOPS LAB</b>         |                         |                         |
| Thu         | OOPS                    | PCS-I                   | PCS-I                   | <b>LIBRARY</b>          |                        | COA                     | DS                      | MEFA                    |
| Fri         | OOPS                    | DS                      | COA                     | MFCS                    |                        | MFCS                    | OOPS                    | <b>SPORTS</b>           |
| Sat         | DS                      | MEFA                    | COA                     | PCS-I                   |                        | OOPS                    | MFCS                    | DS                      |

### Staff Details:

| S. No. | Course Code | Course Name                                        | A                                  | B                                       | C                                    | D                                        |
|--------|-------------|----------------------------------------------------|------------------------------------|-----------------------------------------|--------------------------------------|------------------------------------------|
| 1.     | V20MBT51    | Managerial Economics and Financial Analysis (MEFA) | Mr. K Vinay Kumar                  | Ms. K Suji                              | Ms. K Suji                           | Dr. K Pulla Rao                          |
| 2.     | V20MAT07    | Mathematical Foundation Of                         | Mr. P Umasankar                    | Mr. P Umasankar                         | Ms. Y Divya Vani                     | Ms. Y Divya Vani                         |
| 3.     | V20CST03    | OOPs Through C++ (OOPS)                            | Mr. K Lakshmaji                    | Mr. K Lakshmaji                         | Mr. K Lakshmi Narayana               | Mr. K Lakshmi Narayana                   |
| 4.     | V20CST04    | Data Structures (DS)                               | Dr. Ch Raja Ramesh                 | Dr. Ch Raja Ramesh                      | Mrs. B Sri Ramya                     | Mrs. B Sri Ramya                         |
| 5.     | V20CST05    | Computer Organization and Architecture (COA)       | Mrs. A Leelavathi                  | Mrs. A Leelavathi                       | Mrs. D Anjani Suputhri Devi          | Mrs. D Anjani Suputhri Devi              |
| 6.     | V20CSL03    | OOPs Through C++ Lab (OOPs Lab)                    | Mr. K Lakshmaji / Mr. P            | Mr. K Lakshmaji/ Ms. Y Divya Vani       | Mr. K Lakshmi Narayana /             | Mr. K Lakshmi Narayana / Mrs. P Suneetha |
| 7.     | V20CSL04    | Data Structures Lab (DS Lab)                       | Dr. Ch Raja Ramesh / Mrs. D Santhi | Dr. Ch Raja Ramesh /Mrs. A Leelavathi   | Mrs. B Sri Ramya / Mrs. A Leelavathi | Mrs. B Sri Ramya / Mr. M Bhanu Ranga     |
| 8.     | V20CSL05    | Linux Shell Scripting Lab (LSS Lab)                | Mrs. P Suneetha / Mr. M V V        | Mrs. P Suneetha / Mr. M Bhanu Ranga Rao | Dr. D Jaya Kumari / Mr. G N V        | Dr. D Jaya Kumari / Mrs. P Suneetha      |
| 9.     | V20ENT02    | Professional Communication Skills –I (PCS-I)       | Dr. T Sujani / Mr. G Srinivasarao  | Mr. U Aparanji / Mr. G Srinivasarao     | Mr. M Venkata Ramana / Mrs. Ch       | Mr. K V Ramarao / Mrs. K Radha Madhavi   |

**NOTE:** As Part of Curriculum you have to study **Skill Oriented Course-I** also, It will be conducted **any 1 WEEK** during the semester.

| S.No. | Name of the Lab                    | Lab Venue                                |
|-------|------------------------------------|------------------------------------------|
| 1     | OOPs Through C++ Lab(OOPs Lab)     | EF Codd Lab (B-Block Ground Floor)       |
| 2     | Data Structures Lab(DS Lab)        |                                          |
| 3     | Linux Shell Scripting Lab(LSS Lab) | James Gosling Lab (B-Block Ground Floor) |

  
 Head of the Department  
 Dept. of Computer Science & Engineering  
 Sri Vasavi Engineering College  
 TADEPALLIGUDEM-534 101

# ***COURSE STRUCTURE***

## **III Semester**

### **SEMESTER-III (SECOND YEAR)**

| <b>S.No.</b>  | <b>Code</b> | <b>Name of the Course</b>                   | <b>L</b>  | <b>T</b> | <b>P</b>  | <b>C</b>    |
|---------------|-------------|---------------------------------------------|-----------|----------|-----------|-------------|
| 1             | V20MBT51    | Managerial Economics and Financial Analysis | 3         | -        | -         | 3           |
| 2             | V20MAT07    | Mathematical Foundation Of Computer Science | 3         | -        | -         | 3           |
| 3             | V20CST03    | OOPs Through C++                            | 3         | -        | -         | 3           |
| 4             | V20CST04    | Data Structures                             | 3         | -        | -         | 3           |
| 5             | V20CST05    | Computer Organization and Architecture      | 3         | -        | -         | 3           |
| 6             | V20CSL03    | OOPs Through C++ Lab                        | -         | -        | 3         | 1.5         |
| 7             | V20CSL04    | Data Structures Lab                         | -         | -        | 3         | 1.5         |
| 8             | V20CSL05    | Linux Shell Scripting Lab                   | -         | -        | 3         | 1.5         |
| 9             |             | Skill Oriented Course - I                   | <b>1</b>  | <b>0</b> | <b>2</b>  | <b>2</b>    |
| 10            | V20ENT02    | Professional Communication Skills -I        | 2         | -        | -         | 0           |
| <b>Total:</b> |             |                                             | <b>18</b> | <b>0</b> | <b>11</b> | <b>21.5</b> |

**Total ContactHours:29**

**Total Credits:21.5**



# LESSON PLANS

# **Managerial Economics and Financial Analysis**

**Academic Year: 2022-23**

**Programme: B.Tech**

**Year/ Semester: III**

**Section: A,B,C& D**

**Name of the Course: Managerial Economics and Financial Analysis**

**Course Code: V20MBT51**

## **LESSON PLAN**

**Course Outcomes** (Along with Knowledge Level):

After completion of this course, Student will be able to:

| S. No. | CO. No. | Course Outcome                                                                                                | BTL |
|--------|---------|---------------------------------------------------------------------------------------------------------------|-----|
| 1.     | CO1     | Estimate the demand for a product and the relationship between price and demand                               | K2  |
| 2.     | CO2     | Interpret production concept, least cost combinations and various costs concepts in decision making.          | K3  |
| 3.     | CO3     | Differentiate various Markets and Pricing methods along with Business Cycles.                                 | K2  |
| 4.     | CO4     | Prepare financial statements and its analysis.                                                                | K3  |
| 5.     | CO5     | Assess various investment project proposals with the help of Capital Budgeting techniques for decision making | K3  |

**Text Books:**

1. Dr. N. AppaRao, Dr. P. Vijay Kumar: 'Managerial Economics and Financial Analysis', Cengage Publications, New Delhi – 20112.
2. Dr. A. R. Aryasri – Managerial Economics and Financial Analysis, TMH 2011

**Reference Books:**

1. Dr. B. Kuberudu and Dr. T. V. Ramana: Managerial Economics & Financial Analysis, Himalaya Publishing House, 2014
2. S. A. Siddiqui; A. S. Siddiqui: Managerial Economics and Financial Analysis, New Age International Publishers, 2012.

**Targeted Proficiency and attainment Levels (for each Course Outcome):**

| Cos                          |         | CO1 | CO2 | CO3 | CO4 | CO5 |
|------------------------------|---------|-----|-----|-----|-----|-----|
| Targeted Proficiency Level   |         | 60  | 60  | 60  | 60  | 60  |
| Targeted level of Attainment | Level 3 | 60  | 60  | 60  | 60  | 60  |
|                              | Level 2 | 50  | 50  | 50  | 50  | 50  |
|                              | Level 1 | 40  | 40  | 40  | 40  | 40  |

### Lecture Plan:

| S.No | Course Outcome                                                                   | Intended learning outcomes (ILO)                                     | Knowledge level of ILO | No. of Hours required | Pedagogy        | Teaching aids |
|------|----------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------|-----------------------|-----------------|---------------|
|      | Estimate the demand for a product and the relationship between price and demand. | <b>UNIT-I: Introduction to managerial economics</b>                  |                        | 11                    |                 |               |
| 1    |                                                                                  | Define managerial economics                                          | K1                     | 1                     | Lecture Discuss | Blackboard    |
| 2    |                                                                                  | Describe ME with other disciplines                                   | K1                     | 1                     | Lecture         | Blackboard    |
| 3    |                                                                                  | Explain Nature and scope of managerial economics                     | K2                     | 1                     | Lecture         | Blackboard    |
| 4    |                                                                                  | Define Demand                                                        | K1                     | 1                     | Lecture Discuss | Blackboard    |
| 5    |                                                                                  | Describe law of demand                                               | K2                     | 1                     | Lecture         | Blackboard    |
| 6    |                                                                                  | Explain Elasticity of demand                                         | K2                     | 2                     | Lecture         | Blackboard    |
| 7    |                                                                                  | Find the of elasticity of demand                                     | K2                     | 2                     | Lecture         | Blackboard    |
| 8    |                                                                                  | Explain Demand forecasting, methods.                                 | K2                     | 2                     | Lecture Discuss | Blackboard    |
|      |                                                                                  | <b>UNIT-II: Production analysis and cost analysis</b>                |                        | 8                     |                 |               |
| 1    | Interpret production concept, least cost combinations and various costs concepts | State Production function                                            | K1                     | 1                     | Lecture         | Blackboard    |
| 2    |                                                                                  | State Isocost                                                        | K1                     | 1                     | Lecture         | Blackboard    |
| 3    |                                                                                  | State Iso quants                                                     | K1                     | 1                     | Lecture         | Blackboard    |
| 4    |                                                                                  | Explain Cob-Douglas production function                              | K2                     | 1                     | Lecture Discuss | Blackboard    |
| 5    |                                                                                  | Describe economies of scale                                          | K2                     | 1                     | Lecture         | Blackboard    |
|      |                                                                                  | Enumerate various cost concepts                                      | K1                     | 1                     | Lecture         | Blackboard    |
|      |                                                                                  | Solve break even analysis problems                                   | K3                     | 2                     | Lecture         | Blackboard    |
|      |                                                                                  | <b>UNIT-III: market structures, Forms of Business organizations</b>  |                        | 14                    |                 |               |
| 1    | Differentiate various Markets and Pricing methods along with Business Cycles.    | Describe Different types of market structures                        | K1                     | 3                     | Lecture         | Blackboard    |
| 2    |                                                                                  | Explain Price-output determination under different market structures | K2                     | 4                     | Lecture         | Blackboard    |
| 3    |                                                                                  | Explain Pricing objectives, Cost and demand based Pricing methods    | K2                     | 2                     | Lecture         | Blackboard    |
| 4    |                                                                                  | Describe competition, strategy based pricing methods.                | K2                     | 2                     | Lecture         | Blackboard    |
| 5    |                                                                                  | State the meaning and features of business cycles                    | K1                     | 1                     | Lecture         | Blackboard    |
| 6    |                                                                                  | Describe the Phases of business Cycles.                              | K2                     | 2                     | Lecture         | Blackboard    |



|   |                                                              |                                                          |    |    |         |            |
|---|--------------------------------------------------------------|----------------------------------------------------------|----|----|---------|------------|
|   |                                                              | <b>Unit–IV: financial accounting</b>                     |    | 13 |         |            |
| 1 | Prepare the financial statements to gains                    | Describe double entry system                             | K2 | 3  | Lecture | Blackboard |
| 2 |                                                              | Preparation of financial statements                      | K3 | 4  | Lecture | Blackboard |
| 3 |                                                              | Interpretation of financial statements by using, Ratios. | K3 | 6  | Lecture | Blackboard |
|   |                                                              | <b>UNIT–V: capital budgeting</b>                         |    | 08 |         |            |
| 1 | Assess various investment project proposals with the help of | Define Capital                                           | K1 | 1  | Lecture | Blackboard |
| 2 |                                                              | Enumerate types of capital                               | K1 | 1  | Lecture | Blackboard |
| 3 |                                                              | Explain capital budgeting, Process                       | K2 | 2  | Lecture | Blackboard |
| 5 |                                                              | Apply capital budgeting techniques                       | K3 | 4  | Lecture | Blackboard |

**Total No. of Classes: 54**

# **Mathematical Foundation of Computer Science**

**Academic Year: 2022-23**

**Programme: B.Tech**

**Year/ Semester: III**

**Section: A,B,C& D**

**Name of the Course: Mathematical Foundation of Computer Science**

**Course Code: V20MAT07**

## **LESSON PLAN**

**Course Outcomes (Along with Knowledge Level):**

After completion of this course, Student will be able to:

| S. No | CO.No | Course Outcomes                                                                            | BTL |
|-------|-------|--------------------------------------------------------------------------------------------|-----|
| 1     | CO1   | Demonstrate the concepts associated with propositions and mathematical logic               | K3  |
| 2     | CO2   | Demonstrate the basic concepts associated with relations, functions and their applications | K3  |
| 3     | CO3   | Solve recurrence relations using various methods                                           | K3  |
| 4     | CO4   | Apply techniques of graphs for real-time problems                                          | K3  |
| 5     | CO5   | Construct minimal spanning tree by using different algorithms                              | K3  |

### **TEXT BOOKS:**

1. Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, 1st Edition, Tata McGraw Hill.
2. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
3. Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T.P. Baker, 2nd Edition, Prentice Hall of India.

### **REFERENCE BOOKS:**

1. Elements of Discrete Mathematics -A Computer Oriented Approach, C. L. Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
2. Discrete Mathematics with Combinatorics and Graph Theory, Santha, 1st Edition Cengage Learning.

Targeted Proficiency and attainment Levels (for each Course Outcome):

| Cos                          |         | CO1 | CO2 | CO3 | CO4 | CO5 |
|------------------------------|---------|-----|-----|-----|-----|-----|
| Targeted Proficiency Level   |         | 60  | 60  | 60  | 60  | 60  |
| Targeted level of Attainment | Level 3 | 60  | 60  | 60  | 60  | 60  |
|                              | Level 2 | 50  | 50  | 50  | 50  | 50  |
|                              | Level 1 | 40  | 40  | 40  | 40  | 40  |

**Lecture Plan:****UNIT-1**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                        | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids |
|------|----------------|-----------------------------------------------------------------------------------------|------------------------|-----------------------|-------------------------|---------------|
| 1    | CO1            | Dissemination of Vision, Mission, PEOs, POs, PSOs                                       |                        | 1                     | Lecture                 | ICT           |
| 2    |                | <b>Mathematical Logic:</b><br>Define Statements and their Notations, Connectives        | K1                     | 1                     | Lecture                 | BB            |
| 3    |                | Describe Well Formed Formulas, Truth Tables, Tautologies                                | K2                     | 1                     | Lecture with Discussion | BB            |
| 4    |                | Explain equivalence of Formulas                                                         | K2                     | 2                     | Lecture                 | BB            |
| 5    |                | State duality Law, Tautological implications                                            | K1                     | 1                     | Lecture with Discussion | BB            |
| 6    |                | Explain normal forms                                                                    | K2                     | 2                     | Lecture                 | BB            |
| 7    |                | Illustrate theory of inference for statement calculus                                   | K3                     | 2                     | Lecture                 | BB            |
| 8    |                | Practice indirect method of proof                                                       | K3                     | 2                     | Lecture                 | BB            |
| 9    |                | Identify statement functions, variables and quantifiers, free and bound Variables       | K2                     | 1                     | Lecture                 | BB            |
| 10   |                | Recognize theory for predicate calculus- Predicates, quantifiers, universe of discourse | K2                     | 2                     | Lecture                 | ICT           |
| 11   |                | Total                                                                                   |                        | 15                    |                         |               |

**UNIT-2**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                     | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids |
|------|----------------|--------------------------------------------------------------------------------------|------------------------|-----------------------|-------------------------|---------------|
| 1    | CO 2           | <b>Set Theory and Relations:</b><br>Define basic concepts                            | K1                     | 1                     | Lecture                 | ICT           |
| 2    |                | Illustrate operations on binary sets                                                 | K2                     | 1                     | Lecture                 | ICT           |
| 3    |                | Use principle of inclusion and exclusion                                             | K3                     | 1                     | Lecture                 | ICT           |
| 4    |                | Describe Relation and properties of binary relations on a set and Transitive Closure | K2                     | 1                     | Lecture                 | BB            |
| 5    |                | Sketch out relation matrix and digraph                                               | K3                     | 1                     | Lecture with Discussion | BB            |

|   |  |                                                                     |    |           |                         |    |
|---|--|---------------------------------------------------------------------|----|-----------|-------------------------|----|
| 6 |  | Practice equivalence, and partial ordering relations                | K3 | 2         | Lecture with Discussion | BB |
| 7 |  | Construct hasse diagrams, lattice and state its properties.         | K3 | 2         | Lecture                 | BB |
| 8 |  | Illustrate Functions, bijective functions, composition of functions | K3 | 1         | Lecture                 | BB |
| 9 |  | Total                                                               |    | <b>10</b> |                         |    |

### UNIT-3

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                 | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids |
|------|----------------|------------------------------------------------------------------|------------------------|-----------------------|-------------------------|---------------|
| 1    | CO3            | <b>Recurrence Relations</b><br>Explain Generating of functions   | K2                     | 2                     | Lecture                 | ICT           |
| 2    |                | Calculate Coefficient of generating functions                    | K3                     | 2                     | Lecture                 | BB            |
| 3    |                | Explain Recurrence relations                                     | K2                     | 1                     | Lecture with discussion | BB            |
| 4    |                | Solve homogeneous Recurrence relations by method of substitution | K3                     | 1                     | Lecture                 | ICT           |
| 5    |                | Solve homogeneous Recurrence relations by Generating functions   | K3                     | 2                     | Lecture with discussion | BB            |
| 6    |                | Solve Recurrence relations by method of characteristic roots     | K3                     | 2                     | Lecture with discussion | BB            |
| 7    |                | Solve inhomogeneous recurrence relations                         | K3                     | 3                     | Lecture                 | BB            |
| 8    |                | Total                                                            |                        | <b>13</b>             |                         |               |

**UNIT-4**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                   | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids |
|------|----------------|------------------------------------------------------------------------------------|------------------------|-----------------------|-------------------------|---------------|
| 1    | CO 4           | Describe basic concepts of graphs                                                  | K1                     | 2                     | Lecture with Discussion | ICT           |
| 2    |                | Illustrate matrix representation of graphs, Adjacency matrices, Incidence matrices | K2                     | 1                     | Lecture                 | BB            |
| 3    |                | Find subgraph, isomorphic graphs, paths and circuits                               | K3                     | 2                     | Lecture                 | ICT           |
| 4    |                | Demonstrate Eulerian and Hamiltonian Graphs, Multigraphs                           | K3                     | 2                     | Lecture with Discussion | BB            |
| 5    |                | Use Euler's Formula for Planar Graphs                                              | K3                     | 1                     | Lecture with Discussion | BB            |
| 6    |                | Explain Graph Colouring and Chromatic Number                                       | K2                     | 1                     | Lecture with Discussion | BB            |
| 7    |                | Total                                                                              |                        | 9                     |                         |               |

**UNIT-5**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                         | Knowledge Level of ILO | No. of Hours Required | Pedagogy | Teaching aids |
|------|----------------|----------------------------------------------------------|------------------------|-----------------------|----------|---------------|
| 1    | CO 5           | Explain tree and spanning trees                          | K2                     | 2                     | Lecture  | BB            |
| 2    |                | Sketch Minimal spanning trees using Kruskal's algorithms | K3                     | 2                     | Lecture  | BB            |
| 3    |                | Sketch Minimal spanning trees using Prim's algorithms    | K3                     | 2                     | Lecture  | BB            |
| 4    |                | Construct BFS                                            | K3                     | 2                     | Lecture  | BB            |
| 5    |                | Construct DFS                                            | K3                     | 2                     | Lecture  | BB            |
| 6    |                | Explain binary trees                                     | K2                     | 2                     | Lecture  | BB            |
|      |                | Explain Planar Graphs                                    | K2                     | 1                     | Lecture  | BB            |
| 9    |                | Total                                                    |                        | 13                    |          |               |

**Total No. of Classes: 60**

## **OOPs Through C++**

**Academic Year: 2022-23**

**Programme: B.Tech**

**Year/ Semester: III**

**Section: A,B,C& D**

**Name of the Course: OOPs Through C++**

**Course Code: V20CST03**

### **LESSON PLAN**

**COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

| S.No. | CO No. | Course Outcome                                                                 | BTL |
|-------|--------|--------------------------------------------------------------------------------|-----|
| 1.    | CO1    | Differentiate Procedural Oriented Programming and Object-Oriented Programming. | K2  |
| 2.    | CO2    | Develop programs using Classes and Objects.                                    | K3  |
| 3.    | CO3    | Demonstrate Constructors, destructors & Operator-Overloading.                  | K3  |
| 4.    | CO4    | Construct Classes using inheritance and Exceptions.                            | K3  |
| 5.    | CO5    | Demonstrate Files and Generic Programming.                                     | K3  |

**Text Books:**

1. Programming in C++, Ashok N Kamthane, 2<sup>nd</sup> Edition, Pearson.
2. C++ How to Program, Paul J. Deitel, Harvey Deitel, 6<sup>th</sup> edition, PHI publication.

**Reference Books:**

1. Object Oriented Programming C++, Joyce Farrell, Cengage.
2. Mastering C++, Venugopal, Raj Kumar, Ravi Kumar, TMH.
3. The Complete Reference C++, Herbert Schildt, 4<sup>th</sup> Edition, Mcgraw Hill.
4. Object Oriented Programming With C++, R. Subburaj, Vikas Publishing House.

**Targeted Proficiency and attainment Levels (for each Course Outcome):**

| Cos                          |         | CO1 | CO2 | CO3 | CO4 | CO5 |
|------------------------------|---------|-----|-----|-----|-----|-----|
| Targeted Proficiency Level   |         | 60  | 60  | 60  | 60  | 60  |
| Targeted level of Attainment | Level 3 | 60  | 60  | 60  | 60  | 60  |
|                              | Level 2 | 55  | 55  | 55  | 55  | 55  |
|                              | Level 1 | 50  | 50  | 50  | 50  | 50  |

**Lecture Plan:****UNIT-1**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                       | Knowledge Level of ILO | No. of Hours | Pedagogy            | Teaching aids |
|------|----------------|------------------------------------------------------------------------|------------------------|--------------|---------------------|---------------|
| 1    | CO1            | Define Object-Oriented Programming                                     | K1                     | 1            | Lecture             | BB/ICT        |
| 2    |                | Describe about Data Types, Variables, Constants, Operators.            | K2                     | 1            | Lecture             | BB/ICT        |
| 3    |                | Explain decision Statements & Control Structures with examples         | K2                     | 1            | Lecture             | BB/ICT        |
| 4    |                | Discuss about Arrays, Namespace, Default Arguments, Constant Arguments | K2                     | 1            | Lecture             | BB/ICT        |
| 5    |                | Discuss Parameter Passing Techniques                                   | K2                     | 2            | Lecture             | BB/ICT        |
| 6    |                | Explain about Features of Object Oriented Programming                  | K2                     | 1            | Lecture+ Discussion | BB/ICT        |
| 7    |                | Illustrate Arrays with example programs                                | K2                     | 2            | Lecture             | BB/ICT        |
| 8    |                | Total                                                                  |                        | 9            |                     |               |

**UNIT-2**

|    |     |                                                                                     |    |    |                     |        |
|----|-----|-------------------------------------------------------------------------------------|----|----|---------------------|--------|
| 1  | CO2 | Define class and object with examples                                               | K1 | 2  | Lecture             | BB/ICT |
| 2  |     | Explain about Access specifiers, Scope Resolution Operator, Static Member variables | K2 | 1  | Lecture             | BB/ICT |
| 3  |     | Describe about Static Member Functions, Array of Objects with examples              | K3 | 2  | Lecture+ Discussion | BB/ICT |
| 4  |     | Illustrate Inline Functions, Overloading Member Functions with example programs.    | K3 | 1  | Lecture             | BB/ICT |
| 5  |     | Explain about Objects as Function Arguments with examples.                          | K2 | 1  | Lecture             | BB/ICT |
| 6  |     | Explain about Friend Functions, Friend Class with an example program.               | K2 | 1  | Lecture             | BB/ICT |
| 7  |     | Construct Local Class, Empty Class with an example program.                         | K3 | 2  | Lecture+ Discussion | BB/ICT |
| 8  |     | Construct Nested Classes with an example program and explain.                       | K3 | 1  | Lecture             | BB/ICT |
| 9  |     | Prepare a program using the concept Return by Reference                             | K3 | 1  | Lecture             | BB/ICT |
| 10 |     | Total                                                                               |    | 12 |                     |        |

**UNIT-3**

|    |     |                                                                                   |    |    |                     |        |
|----|-----|-----------------------------------------------------------------------------------|----|----|---------------------|--------|
| 1  | CO3 | Define Constructor                                                                | K1 | 1  | Lecture             | BB/ICT |
| 2  |     | Explain the characteristics of a constructor.                                     | K2 | 1  | Lecture             | BB/ICT |
| 3  |     | Demonstrate about Constructor with default arguments, Parameterized constructors, | K3 | 2  | Lecture             | BB/ICT |
| 4  |     | Illustrate about Overloading constructors, Copy constructors with an example.     | K3 | 1  | Lecture             | BB/ICT |
| 5  |     | Construct Dynamic Constructors and Destructors                                    | K3 | 1  | Lecture             | BB/ICT |
| 6  |     | Explain about Anonymous Objects.                                                  | K2 | 1  | Lecture             | BB/ICT |
| 7  |     | Explain Operator overloading and its rules with an example.                       | K2 | 1  | Lecture             | BB/ICT |
| 8  |     | Demonstrate unary and binary operators                                            | K3 | 2  | Lecture+ Discussion | BB/ICT |
| 9  |     | Demonstrate about this keyword, Constraint on Increment and Decrement Operators   | K3 | 1  | Lecture             | BB/ICT |
| 10 |     | Illustrate Overloading with Friend Functions with an example                      | K3 | 1  | Lecture             | BB/ICT |
| 11 |     | Explain about Type Conversions                                                    | K2 | 1  | Lecture             | BB/ICT |
| 12 |     | Total                                                                             |    | 13 |                     |        |



**UNIT-4**

|    |     |                                                                                                                                                     |    |    |         |        |
|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---------|--------|
| 1  | CO4 | Define inheritance                                                                                                                                  | K1 | 1  | Lecture | BB/ICT |
| 2  |     | Explain about types of inheritances(Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Hybrid Inheritance) | K2 | 1  | Lecture | BB/ICT |
| 3  |     | Illustrate example programs on Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Hybrid Inheritance       | K3 | 3  | Lecture | BB/ICT |
| 4  |     | Demonstrate Virtual Base Classes with example programs.                                                                                             | K3 | 2  | Lecture | BB/ICT |
| 5  |     | Explain about Constructor in Derived Classes. qualifier classes and inheritance.                                                                    | K2 | 2  | Lecture | BB/ICT |
| 6  |     | Explain about Early Vs Late Binding, Pure Virtual Functions, Virtual Destructor                                                                     |    | 1  | Lecture | BB/ICT |
| 7  |     | Define Exception handling                                                                                                                           | K1 | 1  | Lecture | BB/ICT |
| 8  |     | Explain about the keywords in exception handling with an example                                                                                    | K2 | 1  | Lecture | BB/ICT |
| 9  |     | Demonstrate Multiple Catch Statements.                                                                                                              | K3 | 1  | Lecture | BB/ICT |
| 10 |     | Illustrate about Catching Multiple Exceptions with an example.                                                                                      | K3 | 1  | Lecture | BB/ICT |
| 11 |     | Explain about Re-throwing Exception                                                                                                                 | K2 | 1  | Lecture | BB/ICT |
| 12 |     | Total                                                                                                                                               |    | 15 |         |        |

**UNIT-5**

|    |      |                                                                              |    |    |         |        |
|----|------|------------------------------------------------------------------------------|----|----|---------|--------|
| 1  | CO 5 | Define file and file mode parameters.                                        | K1 | 1  | Lecture | BB/ICT |
| 2  |      | Explain about File Opening Modes, File Stream Classes, I/O manipulators      | K2 | 1  | Lecture | BB/ICT |
| 3  |      | Describe about Classes for File Handling                                     | K2 | 1  | Lecture | BB/ICT |
| 4  |      | Explain Sequential Access Files, Random Access Files with suitable examples. | K2 | 1  | Lecture | BB/ICT |
| 5  |      | Explain about Error Handling Functions                                       | K2 | 1  | Lecture | BB/ICT |
| 6  |      | Define Template, Class Template and Function Template.                       | K1 | 2  | Lecture | BB/ICT |
| 7  |      | Demonstrate about Class Templates and function Templates.                    | K3 | 1  | Lecture | BB/ICT |
| 8  |      | Explain about Standard Template Library.                                     | K2 | 2  | Lecture | BB/ICT |
| 9  |      | Explain about Sequential Containers & Associative Containers.                | K2 | 2  | Lecture | BB/ICT |
| 10 |      | Total                                                                        |    | 12 |         |        |

**Total No. of Classes: 61**

# Data Structures

Academic Year: 2022-23

Year/ Semester: III

Name of the Course: Data Structures

Programme: B.Tech

Section: A,B,C& D

Course Code: V20CST04

## LESSON PLAN

**COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

| S. No | CO.No | Course Outcomes                                                                  | BTL |
|-------|-------|----------------------------------------------------------------------------------|-----|
| 1     | CO1   | Illustrate the time and space complexities for searching and sorting algorithms. | K2  |
| 2     | CO2   | Demonstrate linked lists and their applications.                                 | K3  |
| 3     | CO3   | Demonstrate Stacks and Queues.                                                   | K3  |
| 4     | CO4   | Illustrate basic operations on binary trees.                                     | K3  |
| 5     | CO5   | Demonstrate Graphs and their applications.                                       | K3  |

### **Text Books:**

1. Data Structures, algorithms and applications in C, SartajSahni, Universities press, Second Edition.
2. Fundamentals of Data Structures in C, Ellis Horowitz, SartajSahni and Dinesh Mehta, 2nd Edition, Universities Press (India) Pvt. Ltd.

### **Reference Books:**

1. An Introduction to Data Structures with Application, Jean-Paul Tremblay , Paul Sorenson, Second Edition.
2. Fundamentals of Data Structures and algorithms by C V Sastry, Rakesh Nayak, Ch. Raja Ramesh, IK Publications, new Delhi.
3. Data structures using C and C++, Langsam, Augenstein and Tanenbaum, PHI.
4. Problem solving with C++, The OOP, Fourth edition, W.Savitch, Pearson education.

**Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):**

| Course Outcome | Targeted Proficiency Level<br>(% of Marks) | Targeted level of Attainment<br>(% Students) |
|----------------|--------------------------------------------|----------------------------------------------|
| CO1            | 60                                         | 65                                           |
| CO2            | 60                                         | 65                                           |
| CO3            | 60                                         | 65                                           |
| CO4            | 60                                         | 60                                           |
| CO5            | 60                                         | 60                                           |

## LESSON PLAN

### CO1

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                                                                                                         | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1     | CO 1           | Discuss CO's and syllabus of the course                                                                                                                  | K2                     | 1            | Lecture                 | BB            |
| 2     |                | Classify different types of data Structures (Linear data structures and non-linear data structures)                                                      | K2                     | 1            | Lecture with Discussion | BB+ICT        |
| 3     |                | Describe different algorithm notations<br>i. Performance analysis: Space Complexity, time complexity<br>ii. Asymptotic notation: Big O, Omega and Theta. | K1                     | 2            | Lecture                 | BB            |
| 4     |                | Describe different searching algorithms (Linear search, Binary search, Fibonacci Search)                                                                 | K2                     | 3            | Lecture with Discussion | BB            |
| 5     |                | Explain different sorting techniques (Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort, Radix Sort)                                   | K2                     | 8            | Lecture with Discussion | BB            |
| 6     |                | Discuss about hashing                                                                                                                                    | K2                     | 2            | Lecture with Discussion | BB            |

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### CO2

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                                                             | Knowledge Level of ILO | No. of Hours | Pedagogy                                        | Teaching aids |
|-------|----------------|--------------------------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------------------------------|---------------|
| 1     |                | Demonstrate single linked list by node representation and compute different operations on single linked list | K3                     | 4            | Lecture with Discussion and in class Assignment | BB            |
| 2     |                | Develop different operations like insertion, deletion and display on double linked list.                     | K3                     | 4            | Lecture with Discussion and in class Assignment | BB            |
| 3     |                | Demonstrate Circular Linked List                                                                             | K3                     | 1            | Lecture with Discussion                         | BB+ICT        |

## CO3

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                     | Knowledge Level of ILO | No. of Hours | Pedagogy                                        | Teaching aids |
|------|----------------|----------------------------------------------------------------------|------------------------|--------------|-------------------------------------------------|---------------|
| 1    | CO 3           | Describe stack and array representation of stack                     | K2                     | 1            | Lecture with Discussion                         | BB            |
| 2    |                | Describe representation of stack using linked list                   | K2                     | 1            | Lecture with Discussion                         | BB            |
| 3    |                | Illustrate Towers of hanoi with an example                           | K3                     | 2            | Lecture with Discussion and in class Assignment | BB            |
| 4    |                | Compute infix to postfix conversion                                  | K3                     | 2            | Lecture with Discussion and in class Assignment | BB+ICT        |
| 5    |                | Compute expression evaluation                                        | K3                     | 2            | Lecture with Discussion and in class Assignment | BB+ICT        |
| 6    |                | Describe Queue and its representation using array and its operations | K2                     | 2            | Lecture with Discussion and in class Assignment | BB+ICT        |
| 7    |                | Explain representation of queue with linked list and its operations  | K2                     | 2            | Lecture with Discussion                         | BB            |
| 8    |                | Explain different applications of Queues                             | K2                     | 1            | Lecture with Discussion                         | BB            |
| 9    |                | Describe Circular queue and its operations                           | K2                     | 2            | Lecture with Discussion                         | BB            |

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**CO4**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                         | Knowledge Level of ILO | No. of Hours | Pedagogy                                        | Teaching aids |
|------|----------------|------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------------------------------|---------------|
| 1    | CO 4           | Describe Tree and different types of Trees                                               | K1                     | 2            | Lecture                                         | BB+CRE        |
| 2    |                | Explain Binary Tree, properties of Binary Tree and its representation                    | K2                     | 1            | Lecture with Discussion                         | BB            |
| 3    |                | Develop different operations on Binary tree                                              | K3                     | 1            | Lecture with Discussion and in class Assignment | BB            |
| 4    |                | Describe different Tree Traversals (recursive and non-recursive)                         | K1                     | 2            | Lecture with Discussion                         | BB            |
|      |                | Describe Binary Search Tree and its operations                                           | K3                     | 3            | Lecture with Discussion and in class Assignment | BB+ICT        |
| 5    |                | <b>Explain about</b> Heaps, different types of heaps, properties and operations on heaps | K2                     | 1            | Lecture with Discussion                         | BB            |

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**CO5**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                                                     | Knowledge Level of ILO | No. of Hours | Pedagogy                                        | Teaching aids |
|------|----------------|----------------------------------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------------------------------|---------------|
| 1    | CO6            | Describe graph, properties of graph, properties of graph, representation of graph and elementary operations on graph | K1                     | 2            | Lecture                                         | BB            |
| 2    |                | Illustrate the graph traversal using Depth First Search, Breadth First Search techniques.                            | K3                     | 2            | Lecture                                         | BB            |
| 3    |                | Explain about Spanning Trees                                                                                         | K2                     | 1            | Lecture                                         | BB            |
| 4    |                | Construct Kruskal's Algorithm and Prims Algorithm for minimum spanning tree                                          | K3                     | 2            | Lecture                                         | BB            |
| 5    |                | Construct single source shortest Path Algorithm                                                                      | K3                     | 1            | Lecture with Discussion and in class Assignment | BB            |
| 6    |                | Construct All pair shortest path algorithm to find shortest graph distances.                                         | K3                     | 1            | Lecture with Discussion                         | BB            |

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**Total No. of Classes: 60**

# **Computer Organization and Architecture**

Academic Year: 2022-23

Programme: B.Tech

Year/ Semester: III

Section: A,B,C& D

Name of the Course: Computer Organization and Architecture Course Code: V20CST05

## **LESSON PLAN**

**COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

| S.No. | CO No. | Course Outcome                                                                        | BTL |
|-------|--------|---------------------------------------------------------------------------------------|-----|
| 1.    | CO1    | Illustrate Basic structure of Computers, Instruction types and their addressing modes | K2  |
| 2.    | CO2    | Describe the different modes of Input / Output transfer.                              | K2  |
| 3.    | CO3    | Illustrate different types of Memory.                                                 | K2  |
| 4.    | CO4    | Describe the different types of Control Unit techniques                               | K2  |
| 5.    | CO5    | Explain the concept of Pipelining and Parallel processing.                            | K2  |

1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 5th Edition, McGraw Hill Education.
2. Computer System Architecture, M. Morris Mano, 3rd Edition, Pearson Education.
3. David A. Patterson and John L. Hennessy, Computer Organization and Design: The Hardware/Software Interface, Fifth Edition, Morgan Kaufmann / Elsevier, 2014.

## **REFERENCE BOOKS:**

1. Computer Organization and Architecture, William Stallings, 10th Edition, Pearson Education.
2. Computer Architecture and Organization, John P. Hayes, 3rd Edition, McGraw Hill Education.

## **Targeted Proficiency and attainment Levels (for each Course Outcome):**

| Cos                          |         | CO1 | CO2 | CO3 | CO4 | CO5 |
|------------------------------|---------|-----|-----|-----|-----|-----|
| Targeted Proficiency Level   |         | 65  | 65  | 65  | 65  | 65  |
| Targeted level of Attainment | Level 3 | 65  | 65  | 65  | 65  | 65  |
|                              | Level 2 | 60  | 60  | 60  | 60  | 60  |
|                              | Level 1 | 55  | 55  | 55  | 55  | 55  |



### Lecture Plan:

| S.No | Course Outcome                                                                        | Intended Learning Outcomes (ILO)                                                                         | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | <b>CO 1:</b><br><b>Introduction &amp; Instruction Sequencing and Addressing Modes</b> | Dissemination of Vision, Mission of the Dept. and PEOs, Pos, & PSOs of the Programme, CO'S and syllabus. |                        | 1            | Lecture                 | ICT           |
| 2    |                                                                                       | Describe various Functional units of a computer                                                          | K1                     | 1            | Lecture                 | BB            |
| 3    |                                                                                       | Illustrate the Basic Operational concepts                                                                | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 4    |                                                                                       | Discuss the Bus structures of computers.                                                                 | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 5    |                                                                                       | Explain the instructions and instruction sequencing.                                                     | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 6    |                                                                                       | Illustrate the various addressing modes of instructions.                                                 | K2                     | 3            | Lecture with Discussion | BB/ICT        |
| 7    |                                                                                       | Discuss the basic input/output operation of computers.                                                   | K2                     | 2            | Lecture with Discussion | BB/ICT        |

| S.No | Course Outcome                                   | Intended Learning Outcomes (ILO)            | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|--------------------------------------------------|---------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | <b>CO 2:</b><br><b>Input/output Organization</b> | Describe the Accessing Input/output devices | K1                     | 2            | Lecture                 | BB/ICT        |
| 2    |                                                  | Describe the Interrupts                     | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 3    |                                                  | Discuss the Handling Multiple Devices       | K2                     | 1            | Lecture with Discussion | BB/ICT        |
| 4    |                                                  | Explain the Direct Memory Access.           | K2                     | 4            | Lecture with Discussion | BB/ICT        |
| 5    |                                                  | Explain the Buses and its types.            | K2                     | 3            | Lecture with Discussion | BB/ICT        |

| S.No | Course Outcome                       | Intended Learning Outcomes (ILO)   | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|--------------------------------------|------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | <b>CO 3:<br/>Memory Organization</b> | Describe the memory hierarchy      | K1                     | 1            | Lecture                 | BB/ICT        |
| 2    |                                      | Explain the main memory            | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 3    |                                      | Illustrate the auxiliary memory    | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 4    |                                      | Illustrate the Associative memory. | K2                     | 3            | Lecture with Discussion | BB/ICT        |
| 5    |                                      | Illustrate the cache memory        | K2                     | 3            | Lecture with Discussion | BB/ICT        |

| S.No | Course Outcome                   | Intended Learning Outcomes (ILO)                               | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------------------------|----------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | <b>CO 4:<br/>Processing Unit</b> | Describe the Fundamental Concepts in execution of instruction. | K1                     | 1            | Lecture                 | BB/ICT        |
| 2    |                                  | Describe the Execution of a Complete Instruction               | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 3    |                                  | Explain the Multiple-Bus Organization                          | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 4    |                                  | Discuss the Hardwired Control unit                             | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 5    |                                  | Discuss the Micro Programmed Control unit                      | K2                     | 3            | Lecture with Discussion | BB/ICT        |

| S.No | Course Outcome                                | Intended Learning Outcomes (ILO)                                                 | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|-----------------------------------------------|----------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | <b>CO 5:<br/>Pipelining &amp; Parallelism</b> | Describe the basic concepts of pipelining.                                       | K2                     | 1            | Lecture with Discussion | BB/ICT        |
| 2    |                                               | Discuss the data hazards.                                                        | K2                     | 1            | Lecture with Discussion | BB/ICT        |
| 3    |                                               | Discuss the instruction hazards                                                  | K2                     | 1            | Lecture with Discussion | BB+ICT        |
| 4    |                                               | Discuss the parallel processing challenges                                       | K2                     | 1            | Lecture with Discussion | BB/ICT        |
| 5    |                                               | Discuss the Flynn's classification                                               | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 6    |                                               | Discuss the Vector Architectures                                                 | K2                     | 1            | Lecture with Discussion | BB/ICT        |
| 7    |                                               | Discuss the Hardware multithreading                                              | K2                     | 1            | Lecture with Discussion | BB/ICT        |
| 8    |                                               | Discuss the Multi-core processors and other Shared Memory Multiprocessors        | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 9    |                                               | Discuss the Introduction to Graphics Processing Units, Clusters                  | K2                     | 2            | Lecture with Discussion | BB/ICT        |
| 10   |                                               | Discuss the Warehouse Scale Computers and other Message-Passing Multiprocessors. | K2                     | 2            | Lecture with Discussion | BB/ICT        |

**Total No. of Classes: 60**

## **OOPs Through C++ Lab (OOPs Lab)**

**Academic Year: 2022-23**

**Year/ Semester: III**

**Name of the Course: OOPs Through C++ Lab (OOPs Lab)**

**Programme: B.Tech**

**Section: A,B,C& D**

**CourseCode: V20CSL03**

### **LESSON PLAN**

**COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

| S.No. | CO No. | Course Outcome                                                                                | BTL |
|-------|--------|-----------------------------------------------------------------------------------------------|-----|
| 1.    | CO1    | Develop Programs on Classes and Objects.                                                      | K3  |
| 2.    | CO2    | Demonstrate Constructors, Destructors and Operator-Overloading, Inheritance and Polymorphism. | K3  |
| 3.    | CO3    | Develop programs to handle Exceptions & Files.                                                | K3  |
| 4.    | CO4    | Demonstrate Generic Programming.                                                              | K3  |

**Text Books:**

3. Programming in C++, Ashok N Kamthane, 2<sup>nd</sup> Edition, Pearson.

4. C++ How to Program, Paul J. Deitel, Harvey Deitel, 6<sup>th</sup> edition, PHI publication.

**Targeted Proficiency and attainment Levels (for each Course Outcome):**

| Cos                          |         | CO1 | CO2 | CO3 | CO4 |
|------------------------------|---------|-----|-----|-----|-----|
| Targeted Proficiency Level   |         | 75  | 75  | 75  | 75  |
| Targeted level of Attainment | Level 3 | 70  | 70  | 70  | 70  |
|                              | Level 2 | 65  | 65  | 65  | 65  |
|                              | Level 1 | 55  | 55  | 55  | 55  |

Lecture Plan:

| S.NO | Course Outcome | Intended Learning Outcomes (ILO)                                                                                                                             | Knowledge Level of ILO | No. of Hours required | Pedagogy                | Teaching aids |
|------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------|-------------------------|---------------|
| 1    | CO1            | Demonstrate how to debug basic programs using GDB compiler                                                                                                   | K3                     | 1                     | Lecture With Discussion | ICT           |
| 2    | CO1            | Develop programs on control structures                                                                                                                       | K3                     | 1                     | Lecture With Discussion | ICT           |
| 3    | CO1            | Construct programs for following concepts.<br>a) Default Arguments b) Constant Arguments c) Reference Arguments                                              | K3                     | 3                     | Lecture With Discussion | ICT           |
| 4    | CO2            | Construct programs for following concepts.<br>a) Classes & Objects b) Inline functions<br>c) Static Member functions d) Overloading of Member Functions      | K3                     | 3                     | Lecture With Discussion | ICT           |
| 5    | CO2            | Develop programs for following concepts. a) Objects as Function Arguments b) Friend Functions, Friend class<br>c) Local class d) Empty Class& Nested Classes | K3                     | 3                     | Lecture With Discussion | ICT           |
| 6    | CO3            | Develop programs for following concepts.<br>a) Default constructor b) Constructor with arguments c) Copy constructor                                         | K3                     | 3                     | Lecture With Discussion | ICT           |
| 7    | CO3            | Construct programs for following concepts.<br>a) Binary b) Unary c) new d) delete                                                                            | K3                     | 3                     | Lecture With Discussion | ICT           |
| 8    | CO3            | Construct programs for following concepts.<br>a) Single b) Multilevel c) Hierarchical d) Hybrid                                                              | K3                     | 3                     | Lecture With Discussion | ICT           |
| 9    | CO3            | Demonstrate the use of Virtual Functions & Virtual Base class                                                                                                | K3                     | 3                     | Lecture With Discussion | ICT           |
| 10   | CO3            | Develop programs to handle following Exceptions.<br>a) Division-by-zero b) Overflow in an array                                                              | K3                     | 1                     | Lecture With Discussion | ICT           |

|    |     |                                                                                                                          |    |   |                         |     |
|----|-----|--------------------------------------------------------------------------------------------------------------------------|----|---|-------------------------|-----|
| 11 | CO3 | Develop programs for following file handling operations.<br>a) Copying text files b) Displaying the contents of the file | K3 | 3 | Lecture With Discussion | ICT |
| 12 | CO4 | Demonstrate Class template and Function Template.                                                                        | K3 | 3 | Lecture With Discussion | ICT |
| 13 | CO4 | Demonstrate Sequential Containers & Associative Containers                                                               | K3 | 3 | Lecture With Discussion | ICT |

**Total Hours: 33**

# Data Structures Lab

Academic Year: 2022-23

Year/ Semester: III

Name of the Course: Data Structures Lab

Programme: B.Tech

Section: A,B,C& D

CourseCode: V20CSL04

## LESSON PLAN

**COURSE OUTCOMES (Along with Knowledge Level):**

After completion of this course, the students will be able to:

| S. No. | CO No. | Course Outcome                                            | BTL  |
|--------|--------|-----------------------------------------------------------|------|
| 1.     | CO1    | Construct programs on sorting and searching techniques.   | [K3] |
| 2.     | CO2    | Illustrate various operations on Linked Lists.            | [K3] |
| 3.     | CO3    | Develop programs on Stacks,Queues and their applications. | [K3] |
| 4.     | CO4    | Implement various operations on Trees and Graphs          | [K3] |

**Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):**

| Course Outcome | Targeted Proficiency Level<br>(% of Marks) | Targeted level of Attainment<br>(% Students) |
|----------------|--------------------------------------------|----------------------------------------------|
| CO1            | 60                                         | 65                                           |
| CO2            | 60                                         | 65                                           |
| CO3            | 60                                         | 65                                           |
| CO4            | 60                                         | 60                                           |

### CO 1

| S.No   | Course Outcome | Intended Learning Outcomes (ILO)                                                                                                                                               | Knowledge Level of ILO | No. of Hours | Pedagogy            | Teaching aids |
|--------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|---------------------|---------------|
| 1<br>2 | CO1            | Lab1:Practice following sorting techniques<br>(a)Selection sort (b) Quick sort(c) Merge sort<br>Lab2:Practicefollowing searching methods<br>(a)Linear search (b) Binary search | K3                     | 6            | Lecture& Experiment | BB&ICT        |

**CO2**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                                                | Knowledge Level of ILO | No. of Hours | Pedagogy             | Teaching aids |
|------|----------------|-----------------------------------------------------------------------------------------------------------------|------------------------|--------------|----------------------|---------------|
| 1    | CO2            | Lab 3: Develop a program to implement single linked list and its operations. (create, insert, delete, display). | K3                     | 3            | Lecture & Experiment | ICT           |
| 2    |                | Lab 4: Develop a program to implement double linked list and its operations.                                    | K3                     | 3            | Lecture & Experiment | ICT           |

**CO 3**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                                                                                                                               | Knowledge Level of ILO | No. of Hours | Pedagogy            | Teaching aids |
|------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|---------------------|---------------|
| 1    | CO4            | Lab 5: Construct Stack along with their operations using Arrays.                                                                                                                               | K3                     | 6            | Lecture, Experiment | BB&ICT        |
| 2    |                | Lab 6: Construct Queue along with their operations using Arrays.<br>Lab 7: Develop Circular Queue using Arrays.<br>Lab 8: Construct Queue along with their operations using Single Linked List |                        |              |                     |               |

**CO 4**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                                                            | Knowledge Level of ILO | No. of Hours | Pedagogy            | Teaching aids |
|------|----------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|---------------------|---------------|
| 1    | CO4            | Lab 9: Develop a program to implement Binary search Tree and its operations                                                 | K3                     | 8            | Lecture, Experiment | ICT<br>BB&ICT |
| 2    |                | Lab 10: Demonstrate Depth First Search and Breadth First Search Algorithm.                                                  |                        |              |                     |               |
| 3    |                | Lab 11: Develop Minimum Spanning Tree using Prim's Algorithm.<br>Lab 12: Develop Minimum Spanning Tree Kruskal's Algorithm. |                        |              |                     |               |

**Total no of hours:26**



# Linux Shell Scripting Lab

Academic Year: 2022-23

Year/ Semester: III

Name of the Course: Linux Shell Scripting Lab

Programme: B.Tech

Section: A,B,C& D

CourseCode: V20CSL05

## LESSON PLAN

**COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

| S. No | CO. No | Course Outcomes                                                                                  | BTL |
|-------|--------|--------------------------------------------------------------------------------------------------|-----|
| 1     | CO1    | Demonstrate the basic knowledge of Linux commands and utilities by using Linux shell environment | K3  |
| 2     | CO2    | Experiment with the Concept of shell Programming on Files and Directories                        | K3  |
| 3     | CO3    | Experiment with the Concept of shell Programming on File Permissions                             | K3  |
| 4     | CO4    | Experiment with the Concept of shell Programming on Conditional Statements                       | K3  |
| 5     | CO5    | Experiment with the Concept of shell Programming on Looping Statements                           | K3  |

**Targeted Proficiency and Attainment Levels (for each Course Outcome):**

| COs                          |         | CO1 | CO2 | CO3 | CO4 | CO5 |
|------------------------------|---------|-----|-----|-----|-----|-----|
| Targeted Proficiency Level   |         | 75  | 75  | 75  | 75  | 75  |
| Targeted level of Attainment | Level 3 | 70  | 70  | 70  | 70  | 70  |
|                              | Level 2 | 65  | 65  | 65  | 65  | 65  |
|                              | Level 1 | 55  | 55  | 55  | 55  | 55  |

**Lecture Plan:**

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                           | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|-------|----------------|----------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1     | CO 1           | Dissemination of Department Vision, Mission, PEOs, POs, PSOs               | -                      | 1            | Lecture With Discussion | ICT           |
|       |                | Demonstrate Linux Commands overview and how to interact with Putty Server. | K3                     | 2            | Lecture With Discussion | ICT           |
| 2     |                | Experiment basics of Linux Commands.                                       | K3                     | 3            | Lecture With Discussion | ICT           |
| 3     |                | Experiment General Purpose Utilities.                                      | K3                     | 3            | Lecture With Discussion | ICT           |
| 4     |                | Experiment Displaying Commands.                                            | K3                     | 3            | Lecture With Discussion | ICT           |
| 5     |                | Experiment Commands on Filters.                                            | K3                     | 3            | Lecture With Discussion | ICT           |
| 6     |                | Experiment Disk Handling Utilities.                                        | K3                     | 3            | Lecture With Discussion | ICT           |

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                                          | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|-------|----------------|-------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1     | CO 2           | Experiment Directory Handling Commands.                                                   | K3                     | 3            | Lecture With Discussion | ICT           |
| 2     |                | Experiment File Handling Utilities.                                                       | K3                     | 3            | Lecture With Discussion | ICT           |
| 3     |                | Develop a shell script to print the list of all sub directories in the current directory. | K3                     | 3            | Lecture with Discussion | ICT           |

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                                  | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|-------|----------------|-----------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1     | CO 3           | Develop a shell script that Changes Permissions of files in PWD as rwx for users. | K3                     | 3            | Lecture with Discussion | ICT           |

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                                                                                                                                     | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1     | CO 4           | Develop a Shell Program which receives any year from the keyboard and determine whether the year is leap year or not. If no argument is supplied the current year should be assumed. | K3                     | 3            | Lecture with Discussion | ICT           |
| 2     |                | Develop a shell script which takes two file names as arguments-If their contents are same then delete the second file.                                                               | K3                     | 3            | Lecture with Discussion | ICT           |

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                                                                                                                    | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1     | CO 5           | Develop a shell program to<br>a) Display all the words which are entered as command line arguments.<br>b) Print the given number in the reversed order.             | K3                     | 3            | Lecture with Discussion | ICT           |
| 2     |                | Develop a shell script to delete all lines containing the word 'unix' in the files supplied as arguments.                                                           | K3                     | 3            | Lecture with Discussion | ICT           |
| 3     |                | Develop a shell script to print<br>a) First 25 Fibonacci numbers<br>b) The Prime numbers between the specified range.                                               | K3                     | 3            | Lecture with Discussion | ICT           |
| 4     |                | Develop a shell script Menu driven<br>i) contents of /etc/passwd<br>ii) list of users who have currently logged in.<br>iii) present working directory.<br>iv) exit. | K3                     | 3            | Lecture with Discussion | ICT           |

**Total No. of Classes: 48**

# **Professional Communication Skills - I**

Academic Year: 2022-23

Year/ Semester: III

Name of the Course: Professional Communication Skills - I

Programme: B.Tech

Section: A,B,C& D

CourseCode:V20ENT02

## **LESSON PLAN**

**COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

| S. No. | CO No. | Course Outcome                                                                                                                                                                                                                                                       | BTL |
|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1.     | CO1    | Use vocabulary in regular chores of life with accuracy, make meaningful sentences, and describe people and their traits vividly.                                                                                                                                     | K3  |
| 2.     | CO2    | Distinguish between places of pilgrimage and holiday spots; describe incidents, things and process; and frame questions, statements and expressions.                                                                                                                 | K3  |
| 3.     | CO3    | Demonstrate their knowledge of idioms which are similar to those of native speakers while speaking and writing and use phrases clearly and precisely to articulate their views that compare and contrast indianisms with native expressions and avoid common errors. | K3  |
| 4.     | CO4    | Employ the vocabulary of netizens with ease and walk through the letters and emails for effective official correspondence and infer the accurate meaning of the homophones that are often confusing                                                                  | K3  |
| 5.     | CO5    | Summarize their profile; introduce themselves as well as others by incorporating their accomplishments and Sketch stories and anecdotes in an interesting and engaging manner that arouses curiosity of the audience.                                                |     |

**Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):**

| Course Outcome | Targeted Proficiency Level<br>(% of Marks) | Targeted level of Attainment<br>(% Students) |
|----------------|--------------------------------------------|----------------------------------------------|
| 1              | 50                                         | 60                                           |
| 2              | 50                                         | 60                                           |
| 3              | 50                                         | 60                                           |
| 4              | 50                                         | 60                                           |
| 5              | 50                                         | 60                                           |

**CO 1**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                                              | Knowledge Level of ILO | No. of Hours | Pedagogy             | Teaching aids |
|------|----------------|---------------------------------------------------------------------------------------------------------------|------------------------|--------------|----------------------|---------------|
| 1    | CO 1           | Find appropriate vocabulary to describe occupations, kitchen utensils and tools . <b>Ice-breaking session</b> | K3                     | 1            | Lecture& Discussion  | BB/Handout    |
| 2    |                | Find appropriate vocabulary to describe spices, vegetables, flowers, sciences of study and professions        | K3                     | 1            | Lecture& Discussion  | BB/Handout    |
| 3    |                | Find appropriate vocabulary to describe spices, vegetables, flowers, sciences of study and professions        | K3                     | 1            | Lecture & Discussion | BB/ Handout   |
| 4    |                | Tool-based Activity                                                                                           |                        | 1            |                      |               |
| 5    |                | Construct questions , frame statements and use expressions related to vocabulary learnt <b>Activity</b>       | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 6    |                | Select suitable words and expressions to describe people and physical characteristics                         | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 7    |                | Select suitable words and expressions to talk about mental attributes of people and various professions       | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 8    |                | Select suitable words and expressions to talk about mental attributes of people and various professions       | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 9    |                | Construct questions, frame statements and use expressions related to vocabulary learnt. <b>Activity</b>       | K3                     | 1            | Lecture & Discussion | BB/ Handout   |
| 10   |                | Tool-based Activity                                                                                           |                        | 1            |                      |               |

## CO 2

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                                        | Knowledge Level of ILO | No. of Hours | Pedagogy             | Teaching aids |
|------|----------------|---------------------------------------------------------------------------------------------------------|------------------------|--------------|----------------------|---------------|
| 1    | CO 2           | Describe favorite places and famous places using appropriate vocabulary                                 | K2                     | 1            | Lecture& Discussion  | BB/Handout    |
| 2    |                | Describe places of pilgrimage and holiday spots                                                         | K2                     | 1            | Lecture& Discussion  | BB/Handout    |
| 3    |                | Describe places of pilgrimage and holiday spots. <b>Activity</b>                                        | K2                     | 1            | Lecture & Discussion | BB/ Handout   |
| 4    |                | Tool-based Activity                                                                                     |                        | 1            |                      |               |
| 5    |                | Use suitable expressions to describe a thing and an incident                                            | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 6    |                | Use suitable expressions to describe a thing and an event                                               | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 7    |                | Distinguish between the vocabulary and expressions used to describe a process and a recipe.             | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 8    |                | Distinguish between the vocabulary and expressions used to describe a process and a recipe.             | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 9    |                | Construct questions, frame statements and use expressions related to vocabulary learnt. <b>Activity</b> | K3                     | 1            | Lecture & Discussion | BB/ Handout   |
| 10   |                | Tool-based Activity                                                                                     |                        | 1            |                      |               |

## CO 3

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                             | Knowledge Level of ILO | No. of Hours | Pedagogy             | Teaching aids |
|------|----------------|------------------------------------------------------------------------------|------------------------|--------------|----------------------|---------------|
| 1    | CO 3           | Use idioms and phrases in day to day conversation both in written and spoken | K3                     | 1            | Lecture& Discussion  | BB/Handout    |
| 2    |                | Use idioms and phrases in day to day conversation both in written and spoken | K3                     | 1            | Lecture& Discussion  | BB/Handout    |
| 3    |                | Compare and contrast using “as...as” expressions. <b>Activity</b>            | K4                     | 1            | Lecture & Discussion | BB/ Handout   |
| 4    |                | Tool-based Activity                                                          |                        | 1            |                      |               |
| 5    |                | Examine common mistakes in written and spoken.                               | K4                     | 1            | Lecture & Discussion | BB/ Handout.  |

|    |  |                                                         |    |   |                      |              |
|----|--|---------------------------------------------------------|----|---|----------------------|--------------|
| 6  |  | Examine common mistakes in written and spoken.          | K4 | 1 | Lecture & Discussion | BB/ Handout. |
| 7  |  | Activity on common errors                               |    | 1 | Lecture & Discussion | BB/ Handout. |
| 8  |  | Compare and contrast Indianisms with native expressions | K4 | 1 | Lecture & Discussion | BB/ Handout. |
| 9  |  | Compare and contrast Indianisms with native expressions | K4 | 1 | Lecture & Discussion | BB/ Handout  |
| 10 |  | Tool-based Activity                                     |    | 1 |                      |              |

#### CO 4

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                          | Knowledge Level of ILO | No. of Hours | Pedagogy             | Teaching aids |
|------|----------------|---------------------------------------------------------------------------|------------------------|--------------|----------------------|---------------|
| 1    | CO 4           | Find the vocabulary of netizens                                           | K3                     | 1            | Lecture& Discussion  | BB/Handout    |
| 2    |                | Use Acronyms and abbreviations that are often used by netizens.           | K3                     | 1            | Lecture& Discussion  | BB/Handout    |
| 3    |                | Infer the meaning of the homophones that are often confusing.             | K4                     | 1            | Lecture & Discussion | BB/ Handout   |
| 4    |                | Activity related to Homophones                                            |                        | 1            |                      |               |
| 5    |                | Tool-based Activity                                                       |                        | 1            | Lecture & Discussion | BB/ Handout.  |
| 6    |                | Use the correct format for writing both formal and informal letters       | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 7    |                | Organise ideas and draft emails (Business mail)                           | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 8    |                | Organise ideas and draft emails (Project status mails, informative mails) | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 9    |                | <b>Activity</b> (writing practice)                                        |                        | 1            | Lecture & Discussion | BB/ Handout   |
| 10   |                | Tool-based Activity                                                       |                        | 1            |                      |               |

**CO 5**

| S. No | Course Outcome | Intended Learning Outcomes (ILO)                                   | Knowledge Level of ILO | No. of Hours | Pedagogy             | Teaching aids |
|-------|----------------|--------------------------------------------------------------------|------------------------|--------------|----------------------|---------------|
| 1     | CO 5           | Summarize the strengths, weaknesses and goals                      | K5                     | 1            | Lecture& Discussion  | BB/Handout    |
| 2     |                | Summarize personal and academic achievements                       | K5                     | 1            | Lecture& Discussion  | BB/Handout    |
| 3     |                | <b>Activity</b> (Face to face Interviews)                          |                        | 1            | Lecture & Discussion | BB/ Handout   |
| 4     |                | Tool-based Activity                                                |                        | 1            |                      |               |
| 5     |                | Develop Resume                                                     | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 6     |                | Develop Resume(Practice)                                           | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 7     |                | Sketch stories and anecdotes in an interesting and engaging manner | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 8     |                | Sketch stories and anecdotes in an interesting and engaging manner | K3                     | 1            | Lecture & Discussion | BB/ Handout.  |
| 9     |                | <b>Activity</b> (Story narration)                                  | K3                     | 1            | Lecture & Discussion | BB/ Handout   |
| 10    |                | Tool-based Activity                                                |                        | 1            |                      |               |

**Total No. of Classes: 50**